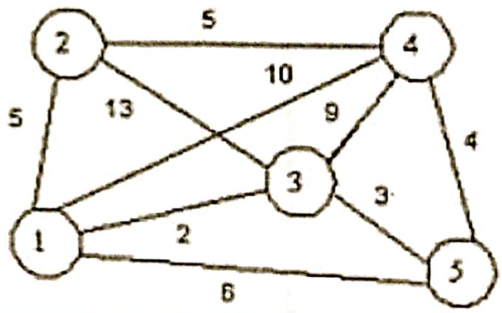
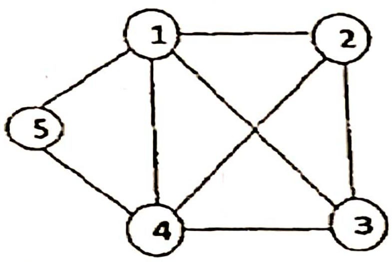
22010170CS

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHAGALPUR
Department of Computer Science and Engineering
CS201 [Design Analysis of Algorithms]
Autumn 2023-2024, END SEMESTER Examination
Time: 3 Hours (03:00 PM to 06:00 PM) Date: 30 November, 2023 Full marks: 50

Sl. No	Instructions: Attempt all the questions. Higher marks will be awarded for concise and to the point answer.	Marks
1	**a.** Find the minimum number of operations required for the following matrix chain multiplication using dynamic programming. A(10x20)*B(20x50)*C(50x1)*D(1x100)	5
	b. Write an algorithm for MST. Estimate the time complexity of this algorithm.	3+2
2	**a.** What are the characteristics of the greedy method?	2
	b. What are the mathematical notations used for algorithm analysis.	3
	c. Find out the shortest path for the following graph using Dijkstra algorithm consider 1 as source node. 	5
3	Write short notes	
	a. Clique decision problem (CDP).	3
	b. 4 queen problems using backtracking with space state tree.	4
	c. Distinguish between backtracking and branch and bound technique.	3
4	**a.** Apply backtracking technique to solve the following graph-coloring problem and also generate the state space tree. 	5
	b. Explain the graph coloring problem and write the algorithm for it.	2+3
5	**a.** Define class P, NP, NP-hard, and NP-complete and also explain their relationship using Venn diagram.	5
	b. What is non deterministic algorithm? Explain with example.	2
	c. Define circuit satisfiability problem and prove that circuit SAT is in class NP.	3